Customer Churn Analysis: Telco

Summary

Of 7,043 customers, **26.5%** churned. This analysis looks at which customer groups leave most often, so the company can focus its retention efforts where they matter most.

Key Findings

1. Contract type is the strongest factor. Month-to-month customers churn at **42.7%**, compared to 11.3% on one-year contracts and just 2.8% on two-year contracts.

2. Fiber optic customers churn at more than double the DSL rate. 41.9% for fiber optic vs 19.0% for DSL and 7.4% for customers with no internet service.

3. Electronic check users are the most likely to leave. 45.3% churn, versus 15–19% for every other payment method.

4. Customers leave early. Churned customers stayed 18.0 months on average, against 37.6 months for those who stayed. They also pay more per month (74.4 vs 61.3).

Recommendations

- Offer discounts or perks to move month-to-month customers onto yearly contracts.
- Review fiber optic pricing and service quality, and follow up with these customers during their first 18 months.
- Encourage electronic check users to switch to automatic payment methods.
- Build an early-warning list: new, month-to-month, fiber, electronic check customers.

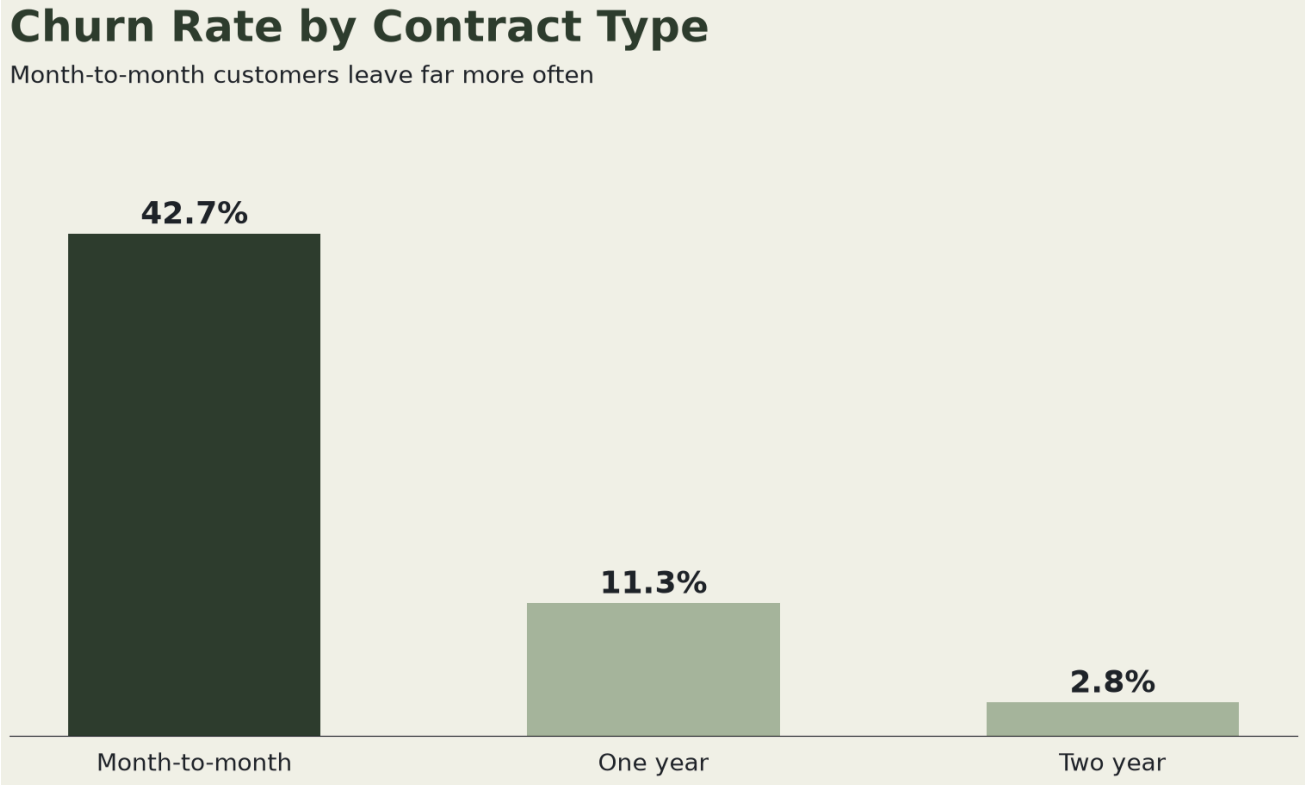
Important Note

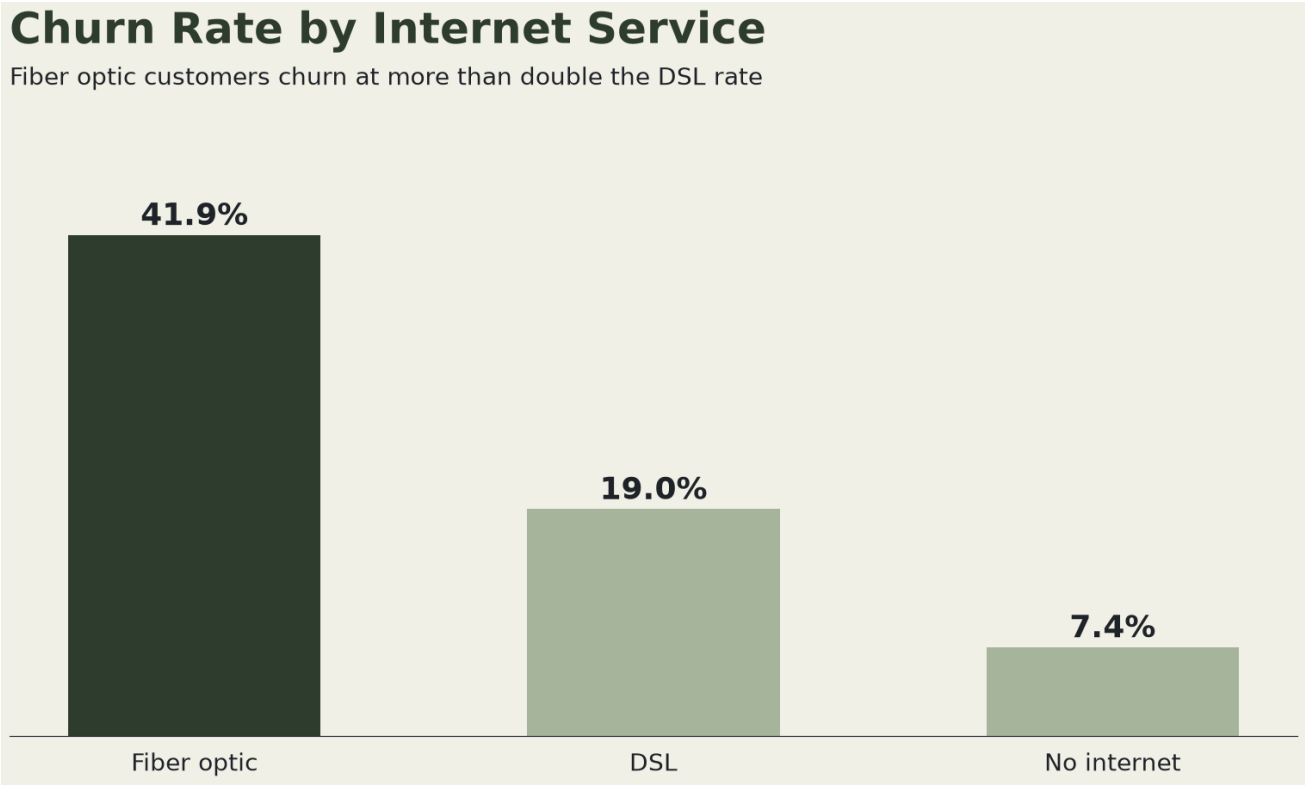
These results show **associations, not proven causes**. For example, fiber optic plans cost more, so the higher churn may be driven by price rather than by the technology itself. Confirming causes would need further analysis or a controlled test.

Data Preparation

- Converted `TotalCharges` to numeric (11 blank values belonged to brand-new customers and were set to 0).
- Unified "No internet service" and "No phone service" into "No".
- Converted `SeniorCitizen` from 0/1 to No/Yes.
- Added a numeric `ChurnFlag` column.

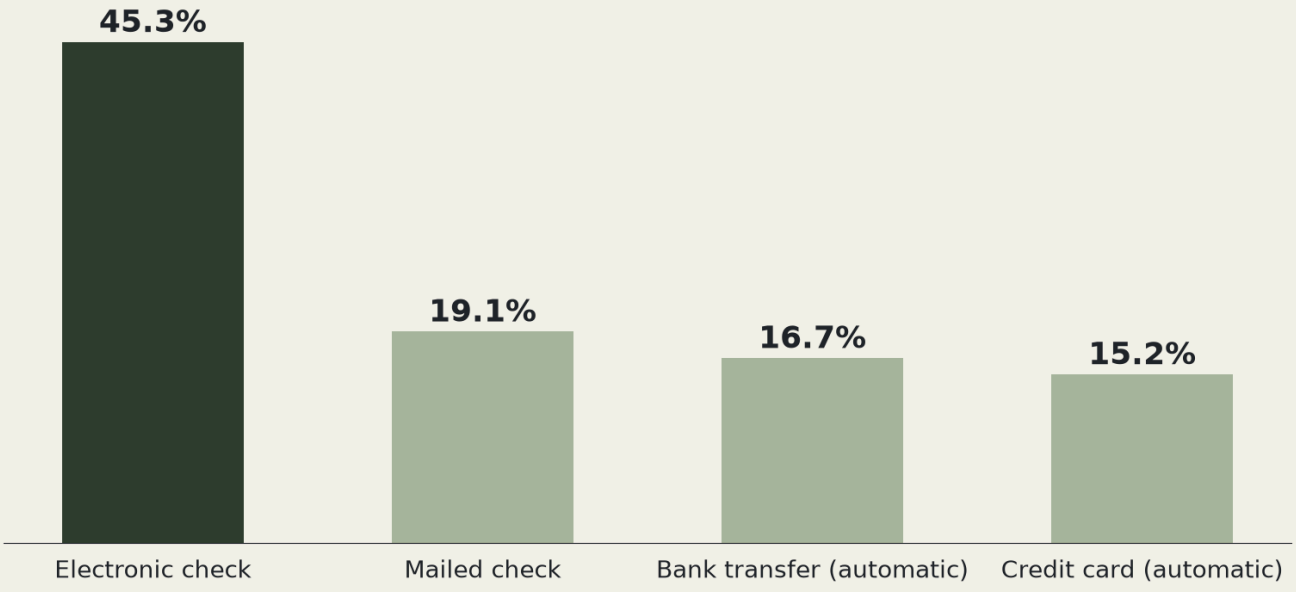
- Final dataset: 7,043 rows, 22 columns, no missing values.





Churn Rate by Payment Method

Electronic check users are the most likely to leave



Average Customer Tenure (months)

Customers who leave do so early

